NETWORKING ASSIGNMENT

WI-FI STANDARDS AND SECURITY

Topics: Wi-Fi standards, home Wi-Fi diagram, network scanning, and SSID hiding

Name: _____

Class / Roll No.: _____

Date: _____

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Task 1: Popular Wi-Fi Standards

QUESTION

List three popular Wi-Fi standards (802.11n, 802.11ac, and 802.11ax) and write one key difference in speed or frequency band for each.

ANSWER

The table below compares three common Wi-Fi standards. Actual speed depends on the router, device, channel width, signal quality, and network conditions.

Wi-Fi standard	Common name	Frequency band	Key speed difference
802.11n	Wi-Fi 4	2.4 GHz and 5 GHz	Theoretical maximum up to 600 Mbps.
802.11ac	Wi-Fi 5	5 GHz	Higher throughput than 802.11n; theoretical maximum varies by configuration and can reach several Gbps.
802.11ax	Wi-Fi 6 / 6E*	2.4 GHz and 5 GHz; 6 GHz with Wi-Fi 6E	Improves efficiency in busy networks and supports higher theoretical speeds, up to about 9.6 Gbps.

*Wi-Fi 6E extends Wi-Fi 6 operation into the 6 GHz band where permitted and supported. Wi-Fi 6 itself commonly operates in 2.4 GHz and 5 GHz bands.

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Task 2: Smartphone Connecting to the Internet

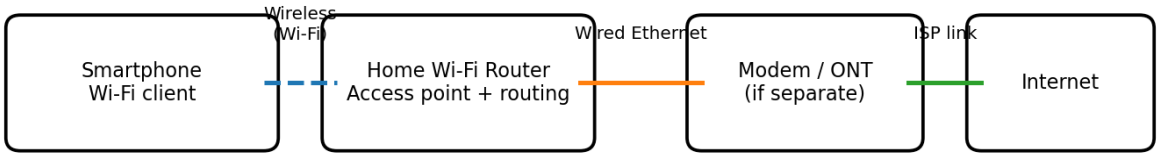
QUESTION

Draw a simple diagram showing how a smartphone connects to the internet through a home Wi-Fi router. Label the wireless and wired parts of the network.

ANSWER

The diagram shows the smartphone connecting wirelessly to the router. The router forwards traffic over a wired Ethernet/ISP connection through a modem or ONT (if separate) to the internet.

Smartphone-to-Internet Home Wi-Fi Diagram



The router sends traffic from the wireless phone to the ISP connection.
If the modem/ONT is built into the router, the separate box is not needed.

Wireless part: smartphone to Wi-Fi router using radio signals. Wired part: router to modem/ONT or ISP equipment using Ethernet or another provider link. Some homes use a combined modem-router, so a separate modem/ONT may not be visible.

Task 3: Scan Nearby Wi-Fi Networks

QUESTION

Scan for available Wi-Fi networks using a laptop or phone. Record the SSID, signal strength, and security type for at least three networks. Explain which one you would choose and why.

ANSWER

Sample observation table (replace these examples with the actual networks visible on your own device; SSIDs, signal strengths, and security settings vary by location):

SSID (example)	Signal strength (example)	Security (example)	Comment
Home_Network	-42 dBm (strong)	WPA2-Personal	Strong signal; suitable if password is secure and router is updated.
Neighbour_WiFi	-67 dBm (medium)	WPA3-Personal	Good security, but signal is weaker and access should only be used with permission.
Guest_WiFi	-78 dBm (weak)	Open / no password	Weak signal and no wireless encryption; avoid for sensitive activity.

I would choose the trusted home network if it has a strong signal, WPA2-AES or WPA3 security, and a strong password. WPA3 is preferable when supported by both the router and device. Signal bars are approximate; dBm values closer to 0 indicate a stronger signal (for example, -42 dBm is stronger than -78 dBm).

For an actual scan: on Windows, select the Wi-Fi icon to view nearby SSIDs and signal bars; use the router's security settings or a trusted network-details tool to confirm security type. On a phone, open Wi-Fi settings and tap a network's details where available. Do not connect to unknown networks or use networks without permission.

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Task 4: SSID Hiding

QUESTION

Explain what SSID hiding is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding a Wi-Fi SSID.

ANSWER

SSID hiding means configuring a Wi-Fi router so it does not openly advertise the network name (SSID) in the usual list of nearby networks. A user may need to enter the network name manually to connect.

Advantage: it can reduce casual visibility of the network name to people browsing nearby Wi-Fi lists.

Disadvantage: it does not provide real security because the network can still be discovered with suitable tools. It can also make connecting new devices less convenient. Strong WPA2-AES/WPA3 security and a strong password are much more important than hiding the SSID.

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